Differentiability

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Differentiability

DIFFERENTIABILITY/DERIVABILITY :

Two fold meaning of derivability

Geometrical meaning of derivative

Slope of the tangent drawn to the curve at x = a if it exists

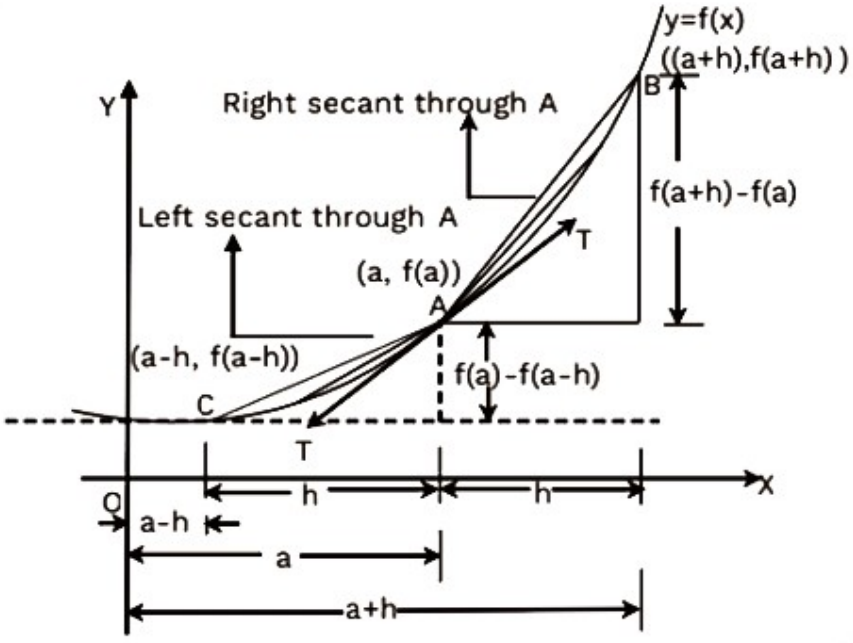
Physical meaning of derivative

(functions which are differentiable) Instantaneous rate of change of function

Note:

"Tangent at a point 'A' is the limiting case of secant through A."

RIGHT AND LEFT HAND DERIVATIVES :



EXISTENCE OF DERIVATIVE :

Right hand & Left hand Derivatives

By definition : $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$

if it exist.

(a) The right hand derivative of f at x = a is denoted by f'(a⁺) is defined by :

$$f'(a^+) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$, provided the limit exists and is finite.

when h → 0, the point B moving along the curve tends to A, i.e., B → A then the chord AB approaches the tangents line AT at the point A.

$$\Rightarrow f'(a^+) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

(b) The left hand derivative of f at x = a denoted by f'(a⁻) is defined by:

$$f'(a^-) = \lim_{h \to 0} \frac{f(a-h) - f(a)}{-h}$$, Provided the limit exists and is finite.

When h → 0, the point C moving along the curve tends to A, i.e., C → A then the chord CA approaches the tangent line AT at point A then

$$\Rightarrow f'(a^-) = \lim_{h \to 0} \frac{f(a-h) - f(a)}{-h}$$

f is said to be derivable at x = a if f'(a⁺) = f'(a⁻) = a finite quantity.

This geometrically means that a unique tangent with finite slope can be drawn at x = a as shown in the figure.

Theorem :

If a function f is derivable at x = a then f is continuous at x = a.

Proof :

Let f is derivable at x = a. Hence

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$ exists.

Also, $f(a+h)-f(a)=\frac{f(a+h)-f(a)}{h}.h \quad (h\neq 0)$

Therefore

$$\lim_{h\to 0}\left[\left(f(a+h)-f(a)\right)\right]=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}.h$$

$= f'(a).0 = 0$

Therefore $\lim_{h\to 0}\left[\left(f(a+h)-f(a)\right)\right]=0$

$$\Rightarrow \lim_{h\to 0} f(a+h)=f(a)$$

$\Rightarrow$ f is continuous at x = a.

$$\Rightarrow \lim_{h\to 0} f(a+h)=f(a)=\lim_{h\to 0} f(a-h)$$

Note :

If f(x) is derivable for every point of its domain of definition, then it is continuous in that domain.

The Converse of the above result is not true:

For a function f :

Differentiability $\Rightarrow$ Continuity;

Continuity $\not\Rightarrow$ derivability.

Non derivability $\not\Rightarrow$ discontinuous

But discontinuity $\Rightarrow$ Non derivability

Note :

(a) Let $f'(a^+) = p$ and $f'(a^-) = q$ where p and q are finite then :

(i) $p = q \Rightarrow$ f is derivable at x = a

$\Rightarrow$ f is continuous at x = a.

(ii) $p \neq q \Rightarrow$ f is not derivable at x = a.

It is very important to note that f may be still continuous at x = a.

(b) If a function f is not differentiable but is continuous at x = a it geometrically implies a sharp corner at x = a.

Q.1 **Find the left and right hand derivative of the following function at given point**

(i) $f(x) = |\ln x|$ at $x = 1$

Sol. $$f(x) = \begin{cases} \ln x, & x \geq 1 \\ -\ln x, & 0 < x \leq 1 \end{cases}$$

$\Rightarrow$ f(x) is continuous at x = 1

$$f'(1^+) = \lim_{h\to 0} \frac{f(1+h) - f(1)}{h} = \lim_{h\to 0} \frac{\ln(1+h) - \ln 1}{h} = \lim_{h\to 0} \frac{\ln(1+h)}{h} = 1$$

$$f'(1^-) = \lim_{h\to 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h\to 0} \frac{-\ln(1-h) - 0}{-h} = \lim_{h\to 0} \frac{\ln(1-h)}{h} = -1$$

Q. **(ii) $f(x) = \ln^2 x$ at $x = 1$**

Sol. $f(x) = \ln^2 x$ is continuous and differentiable function.

$$f'(x) = (2\ln x)\left(\frac{1}{x}\right)$$

$$f'(1) = 2\ln 1\left(\frac{1}{1}\right) = 0$$

Q. **(iii) $f(x) = e^{-|x|}$ at $x = 0$**

Sol. f(x) is continuous at x = 0.

$$f'(0^+) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \frac{e^{-h} - e^0}{h} = -1$$

$$f'(0^-) = \lim_{h\to 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h\to 0} \frac{e^{-h} - e^0}{-h} = 1$$

$\Rightarrow$ f(x) is not differentiable at x = 0.

Q.2 **Discuss the differentiability of $f(x) = \begin{cases} \frac{\sin x^2}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ at $x = 0$**

Sol. For continuity, $\lim_{x\to 0} f(x) = \lim_{h\to 0} \frac{\sinh^2}{h^2}(h)$

$$= \lim_{h\to 0} \frac{\sinh^2}{h^2}(h) = 0$$

Hence, f(x) is continuous at x = 0.

Also, $f'(0^+) = \lim_{h\to 0} \frac{f(h)-f(0)}{h} = \lim_{h\to 0} \frac{\sinh^2}{h^2} = 1$

and $f'(0^-) = \lim_{h\to 0} \frac{f(-h)-f(0)}{-h} = \lim_{h\to 0} \frac{\sinh^2}{h^2} = 1$

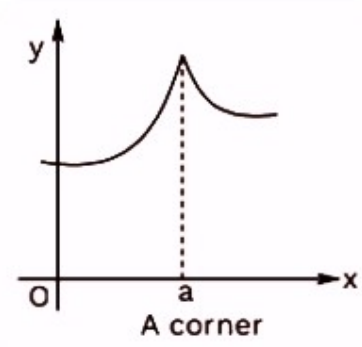
Thus, f(x) is differentiable at x = 0.

How can a function fail to be differentiable :

The function f(x) is said to be non-differentiable at x = a if

(a) Both left and right hand derivative exists but are not equal

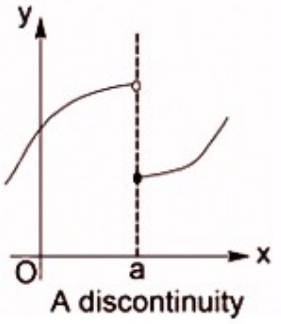
The function y = |x| is not differentiable at 0 as its graph changes direction abruptly when x = 0. In general, if the graph of a function has a 'corner' or 'kink' in it, then the graph of f has no tangent at this point and f is not differentiable there.
While computing f'(a) we find that the left and right limits are different.



A corner

(b) Function is discontinuous at x = a

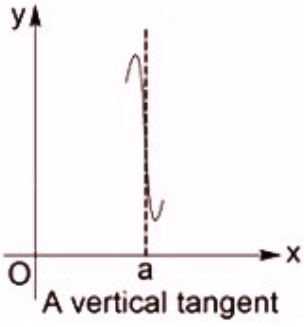
If f is not continuous at x = a then f is not differentiable at a. So at any discontinuity (for instance, a jump of discontinuity) f fails to be differentiable.



A discontinuity

(c) Either or both left and right hand derivative are not finite.
A third possibility is that the curve has a vertical tangent line when x = a, that is f is continuous at a and $\lim_{x\to a} |f'(x)| = \infty$.

This means that the tangent lines become steeper and steeper as x → a.



A vertical tangent

Derivability over An Interval :

f(x) is said to be derivable over an open interval (a, b) if it is derivable at each & every point of the interval.

f(x) is said to be derivable over the closed interval [a, b] if :

Q.3 **Check the differentiability of f(x) = cosx + |cosx| at $x = \frac{\pi}{2}$.**

Sol. $f(x) = \cos x + |\cos x| = \begin{cases} 2\cos x, 0 \le x \le \frac{\pi}{2} \\ 0, \quad \frac{\pi}{2} < x < \pi \end{cases}$

f(x) is continuous at $x = \frac{\pi}{2}$

$$f'\left(\frac{\pi^-}{2}\right) = \lim_{h\to 0} \frac{f\left(\frac{\pi}{2} - h\right) - f\left(\frac{\pi}{2}\right)}{-h} = \lim_{h\to 0} \frac{2\cos\left(\frac{\pi}{2} - h\right) - 2\cos\frac{\pi}{2}}{h} = \lim_{h\to 0} \frac{2\sin h}{-h} = -2$$

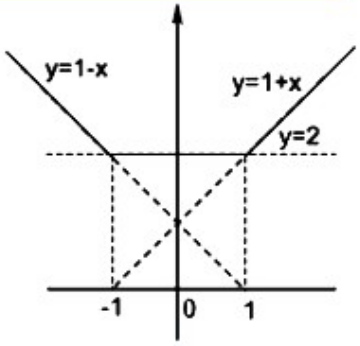
$$f'\left(\frac{\pi^+}{2}\right) = \lim_{h\to 0} \frac{f\left(\frac{\pi}{2} + h\right) - f\left(\frac{\pi}{2}\right)}{h} = \lim_{h\to 0} \frac{0 - 0}{h} = 0$$

$\Rightarrow$ f(x) is not differentiable at $x = \frac{\pi}{2}$

Q.4 **If f(x) = max.{(1–x), (1 + x), 2} then find points where f is not differentiable.**

Sol. f(x) = max. {(1–x), (1 + x), 2}

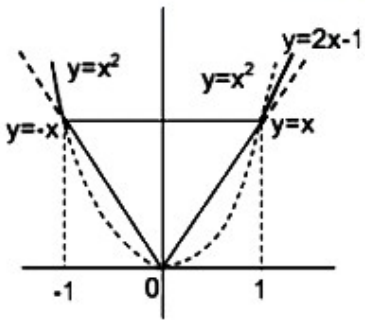
$\Rightarrow$ f(x) is not differentiable at $x = \pm 1$.



Q.5 **Find the points where function** $f(x) = \begin{bmatrix} \max.(|x|, x^2) & \text{if } -\infty < x < 1 \\ \min.(2x-1, x^2) & \text{if } x \ge 1 \end{bmatrix}$ **is not derivable.**

Sol. $f(x) = \begin{bmatrix} \max.(|x|, x^2) & \text{if } -\infty < x < 1 \\ \min.(2x-1, x^2) & \text{if } x \ge 1 \end{bmatrix}$

$\Rightarrow$ f(x) is not differentiable at x = ± 1, 0.



Q.6 **Let f be differentiable at x = a and let f(a) ≠ 0. Evaluate** $\lim_{n\to\infty}\left\{\frac{f\left(a+\frac{1}{n}\right)}{f(a)}\right\}^n$

Sol. $\lim_{n\to\infty}\left\{\frac{f\left(a+\frac{1}{n}\right)}{f(a)}\right\}^n = e^{\lim_{n\to\infty}\left\{\frac{f\left(a+\frac{1}{n}\right)}{f(a)}-1\right\}n} = e^{\lim_{n\to\infty}\left\{\frac{f\left(a+\frac{1}{n}\right)-f(a)}{\left(\frac{1}{n}\right)}\right\}\cdot\frac{1}{f(a)}} = e^{\frac{f'(a)}{f(a)}}$

Q.7 **If** $f(x)=\begin{cases}x^m.\sin\left(\frac{1}{x}\right) & \text{if } x>0\\ 0 & \text{if } x=0\end{cases}$ **is continuous but not differentiable at x = 0 then find m.**

Sol. $f(x)=\begin{cases}x^m.\sin\left(\frac{1}{x}\right) & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$

For continuity at x = 0.

$\lim_{h\to 0} f(0+h)=f(0)\Rightarrow\lim_{h\to 0} h^m\sin\left(\frac{1}{h}\right)=0\Rightarrow m>0$

For function to be not differentiable at x = 0

$\lim_{h\to 0}\frac{f(0+h)-f(0)}{h}=\text{DNE}$

$\lim_{h\to 0}\frac{h^m\sin\left(\frac{1}{h}\right)-0}{h}=\lim_{h\to 0}h^{m-1}\sin\left(\frac{1}{h}\right)=\text{DNE}$

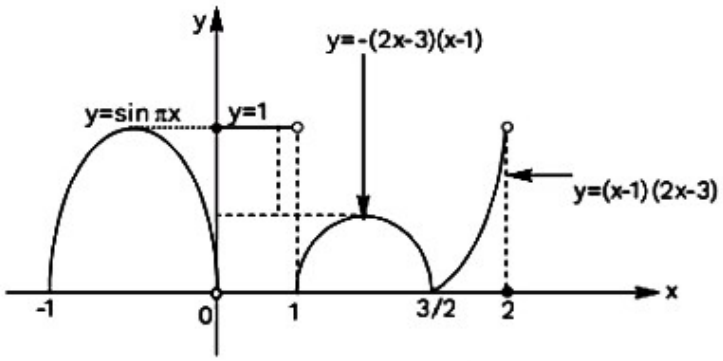
$\Rightarrow m-1\le 0\Rightarrow m\le 1.$

$\therefore\ m\in(0,1]$ for function f(x) to be continuous and not differentiable at x = 0.

Q.8 $f(x)=\begin{cases}\sqrt{4x^2-12x+9}.\{x\}; x\geq 1\\ \cos\left(\frac{\pi}{2}(|x|-\{x\})\right); x<1\end{cases}$ **check the differentiability in $x \in [-1, 2]$.**

Sol. $f(x)=\begin{cases}\sqrt{4x^2-12x+9}.\{x\}; x\geq 1\\ \cos\left(\frac{\pi}{2}(|x|-\{x\})\right); x<1\end{cases}=\begin{cases}|2x-3|.\{x\}, & x\geq 1\\ \cos\frac{\pi}{2}(|x|-\{x\}), & x<1\end{cases}$

$$=\begin{cases}-(2x-3)(x-1), & 1\leq x\leq \frac{3}{2}\\ (2x-3)(x-1), & \frac{3}{2}<x<2\\ \cos\frac{\pi}{2}(-x-(x+1)), & -1\leq x<0\\ \cos\frac{\pi}{2}(x-(x)), & 0\leq x<1\\ 0 & x=2\end{cases}=\begin{cases}-\sin\pi x, & -1\leq x<0\\ 1, & 0\leq x<1\\ -(2x-3)(x-1), & 1\leq x<\frac{3}{2}\\ (2x-3)(x-1), & \frac{3}{2}\leq x<2\\ 0 & x=2\end{cases}$$



$\Rightarrow$ f(x) is not differentiable at $x = 0, 1, \frac{3}{2}, 2$

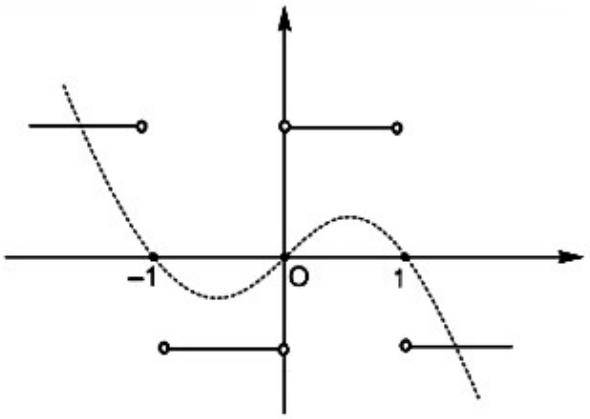
Q.9 **Let $f(x) = \text{sgn}(x)$ and $g(x) = x(1 - x^2)$. Investigate the composite functions $f(g(x))$ and $g(f(x))$ for continuity and differentiability.**

Sol. $f(x) = \text{sgn}(x)$

$$g(x) = x(1-x^2)$$

$$f(g(x)) = \text{sgn}(x(1-x^2))$$

$\Rightarrow$ $fog(x)$ is not differentiable at $x = 0, \pm 1$.



$$\text{as } \text{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

$\therefore$ gof $(x) = 0 \ \forall \ x \in R$

$\Rightarrow$ gof (x) is always differentiable.

$f(g(x))$ is discontinuous & non derivable at -1, 0, 1.

Q.10
$$f(x) = \begin{cases} (\ln(e\,|[x]+[-x]|)^x) \cdot \left(\dfrac{2e^{\left(\frac{\{x\}+\{-x\}}{|x|}\right)} - 5}{3 + e^{\frac{1}{|x|}}} \right) & \text{for } x < 0 \\ 0 & \text{for } x = 0 \\ x \cdot \dfrac{1 - e^{|x|+\{x\}}}{|x| + \{x\}} & \text{for } x > 0 \end{cases}$$

Where [ ], { } represents integral and fractional part functions respectively. Compute the Right hand derivative and Left hand derivative at $x = 0$ and comment on the continuity and derivability at $x = 0$.

Sol. $[x] + [-x] = -1 + 0 = -1$ when $x < 0$ and $x \neq I$

As $\{x\} + \{-x\} = x - [x] + (-x - [-x]) = -([x] + [-x]) = -1(-1 + 0)$

where $x < 0$ and $x \neq I$

$$f(x) = \ln\left(e|-1|\right)^x \cdot \frac{\left(2e^{\frac{1}{|x|}} - 5\right)}{\left(3 + e^{\frac{1}{|x|}}\right)} \quad \text{for } x < 0$$

$$= \frac{x\left(1 - e^{x+x}\right)}{(x+x)} \text{ for } x > 0 \qquad \left[\because |x| = x \ \forall \ x > 0 \text{ and } \{x\} = x \ \forall \ 0 < x < 1\right]$$

$f(x) = 0$ at $x = 0$

$$= x\frac{\left(2e^{\frac{-1}{x}} - 5\right)}{\left(3 + e^{\frac{-1}{x}}\right)} \text{ for } x < 0$$

$$= x\frac{\left(1 - e^{2x}\right)}{2x} \text{ for } x > 0$$

$f(0^-) = f(0^+) = f(0) = 0$

$\Rightarrow f(x)$ is continous at $x = 0$

$$\text{L.H.D.} = \lim_{h\to 0}\frac{f(0-h) - f(0)}{-h} = \lim_{h\to 0}\frac{(-h)}{(-h)}\frac{\left(2e^{\frac{1}{h}} - 5\right)}{\left(3 + e^{\frac{1}{h}}\right)} = \lim_{h\to 0}\frac{\left(2 - 5e^{\frac{-1}{h}}\right)}{\left(3e^{-\frac{1}{h}} + 1\right)} = 2$$

$$\text{R.H.D.} = \lim_{h\to 0}\frac{f(0+h) - f(0)}{h} = \lim_{h\to 0}\frac{\left(1 - e^{2h}\right)}{2h} = -1$$

$f(x)$ is continuous at $x = 0$ but not differentiable at $x = 0$

Q.11 $f(x) = \begin{cases} x + a & \text{if } x < 0 \\ |x - 1| & \text{if } x \geq 0 \end{cases}$ and $g(x) = \begin{cases} x + 1 & , \text{ if } x < 0 \\ (x-1)^2 + b, & \text{if } x \geq 0 \end{cases}$

Where a and b are non-negative real numbers. Determine the composite function gof. If (gof) (x) is continuous for all real x, determine the values of a and b. Further, for these values of a and b, is gof differentiable at $x = 0$? Justify your answer.

Sol. $f(x) = \begin{cases} x + a & \text{if } x < 0 \\ |x - 1| & \text{if } x \geq 0 \end{cases} \Rightarrow f(x) = \begin{cases} x + a & \text{if } x < 0 \\ -(x-1) & \text{if } 0 \leq x < 1 \\ (x-1) & \text{if } 1 \leq x < \infty \end{cases}$

$$g(x) = \begin{cases} x + 1, & \text{if } x < 0 \\ (x-1)^2 + b, & \text{if } x \geq 0 \end{cases}$$

$g(f(x)) = f(x) + 1$, if $f(x) < 0$

$= (f(x) - 1)^2 + b,$

if $f(x) \geq 0$.

So $g(f(x)) = x + a + 1$ if $x + a < 0$ and $x < 0 \Rightarrow x < -a$

$= -x + 1 + 1$ if $-x + 1 < 0$ and $0 \le x < 1 \Rightarrow$ Null set

$= x$ if $x - 1 < 0$ and $1 \le x < \infty \Rightarrow$ Null set

$= (x + a - 1)^2 + b$ if $x + a \ge 0$ and $x < 0 \Rightarrow -a \le x < 0$

$= (-x)^2 + b$ if $-x + 1 \ge 0$ and $0 \le x < 1 \Rightarrow 0 \le x < 1$

$= (x - 2)^2 + b$ if $x - 1 \ge 0$ and $1 \le x < \infty \Rightarrow 1 \le x < \infty$

$g(f(x)) = x + a + 1$ if $-\infty < x < -a$ ...(1)

$= (x + a - 1)^2 + b$ if $-a \le x < 0$ ...(2)

$= x^2 + b$ if $0 \le x < 1$ ...(3)

$= (x - 2)^2 + b$ if $1 \le x < \infty$ ...(4)

g(f(x)) is continuous everywhere so it must be continuous at x =–a, 0, 1.
from (1) and (2), checking continuity at x = –a
$1 = 1 + b \Rightarrow b = 0$
Similarly checking continuity at x = 0,

$(0 + a - 1)^2 + b = 0 + b \Rightarrow a = 1$

$g(f(x)) = x + 2$ if $-\infty < x < -1$
$= x^2$ if $-1 \le x < 0$
$= x^2$ if $0 \le x < 1$
$= (x-2)^2$ if $1 \le x < \infty$

so $(gof)'(x) = 2x$ at $x = 0 \Rightarrow (gof)'(0) = 0$
Hence gof is differentiable at x = 0.

THEOREM :

If f(x) and g(x) both are derivable at x = a, then

(i) f(x) ± g(x) will be differentiable at x = a.

(ii) f(x).g(x) will be differentiable at x = a.

(iii) $\frac{f(x)}{g(x)}$ will be differentiable at x = a if

$g(a) \neq 0$.

Note :

(1) If f(x) is derivable at x = a and g(x) is not derivable at x = a then f(x) + g(x) or f(x) – g(x) will not be derivable at x = a.

e.g. $f(x) = \cos|x|$ is derivable at $x = 0$ and $g(x) = |x|$ is not derivable $x = 0$,
then $\cos|x| + |x|$ or $\cos|x| - |x|$ will not be derivable at $x = 0$.
However nothing can be said about the product function in this case.
$f(x) = x$ derivable at $x = 0$
$g(x) = |x|$ not derivable at $x = 0$

$$fg(x) = x|x| = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x < 0 \end{cases}$$

$f.g(x)$ is derivable at $x = 0$.

(2) If both $f(x)$ and $g(x)$ are non-derivable then nothing definite can be said about the sum/ difference/product function.
$f(x) = \sin|x| - |x|$ is derivable at $x = 0$
$g(x) = \sin|x| + |x|$ is not derivable at $x = 0$.

(3) If $f(x)$ is derivable at $x = a$ and $f(a) = 0$ and $g(x)$ is continuous at $x = a$ then the product function

$F(x) = f(x).g(x)$ will be derivable at $x = a$

$$F'(a^+) = \lim_{h\to 0} \frac{f(a+h)g(a+h) - 0}{h} = f'(a).g(a)$$

$$F'(a^-) = \lim_{h\to 0} \frac{f(a-h)g(a-h) - 0}{-h} = f'(a).g(a)$$

(4) Derivative of a continuous function need not be a continuous function

$$f(x) = \begin{cases} x^2 \sin\frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases} \quad ; \quad \begin{matrix} f'(0^+) = 0 \\ f'(0)^- = 0 \end{matrix}$$

$$f'(x) = \begin{cases} \sin\frac{1}{x}.2x - x^2\left(\cos\frac{1}{x}\right)\left(\frac{-1}{x^2}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

$f'(x)$ is not continuous at $x = 0$.

Q.1 **Let f be defined as follows :** $f(x)=\begin{cases}\sin x & \text{if } x<\pi\\ mx+n & \text{if } x\geq\pi\end{cases}$

where m and n are constants. Determine m and n such that f is derivable on set of real numbers.

Sol. $f(x)=\begin{cases}\sin x, & \text{if } x<\pi\\ mx+n, & \text{if } x\geq\pi\end{cases}$

Since f(x) is continuous at $x=\pi$

$0=m\pi+n$

Now f(x) is differentiable at $x=\pi$

$\cos x\,|_{x=\pi}=m=-1\Rightarrow n=\pi$

Q.2 **Check the differentiability of function at x = e.**

$$f(x)=\begin{cases}(x-e)2^{-2\left(\frac{1}{e-x}\right)}, & x\neq e\\ 0, & x=e\end{cases}$$

Sol. $$f(x)=\begin{cases}(x-e)2^{-2\left(\frac{1}{e-x}\right)}, & x\neq e\\ 0, & x=e\end{cases}$$

$$\text{R.H.D}=\lim_{h\to0}\frac{f(e+h)-f(e)}{h}=\lim_{h\to0}\frac{(h)2^{\frac{2}{h}}}{h}=\infty$$

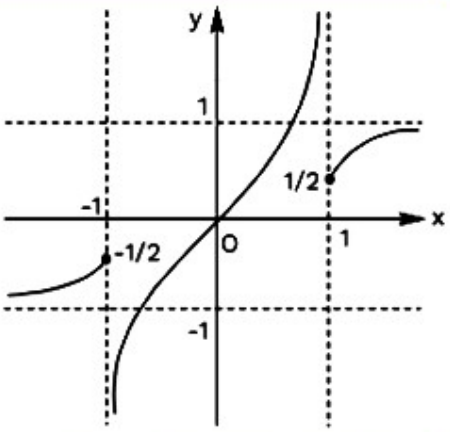
$$\text{L.H.D}=\lim_{h\to0}\frac{f(e-h)-f(e)}{-h}=\lim_{h\to0}\frac{(-h)2^{\frac{-2}{h}}}{-h}=0$$

$\Rightarrow$ f(x) is not differentiable at x = 0.

Q.3 **Find the domain of** $f'(x)$ **if** $f(x)=\begin{cases}\frac{x}{1+|x|} & \text{if } |x|\geq1\\ \frac{x}{1-|x|} & \text{if } |x|<1\end{cases}$

Sol. $f(x)=\begin{cases}\frac{x}{1-x}; -\infty < x \le -1\\ \frac{x}{1+x}; -1 < x < 0\\ \frac{x}{1-x}; \ 0 \le x < 1\\ \frac{x}{1+x}; \ 1 \le x < \infty\end{cases}$

So domain of f'(x) wii be R~{−1, 1}



Q.4 **If f(x) is differentiable at x = a and $f'(a)=\frac{1}{4}$. Find $\lim_{h\to 0}\frac{f(a+2h^2)-f(a-2h^2)}{h^2}$**

Sol. $\text{Limit}=\lim_{h\to 0}\frac{\left(f(a+2h^2)-f(a-2h^2)\right)}{h^2}$ put $t=h^2$

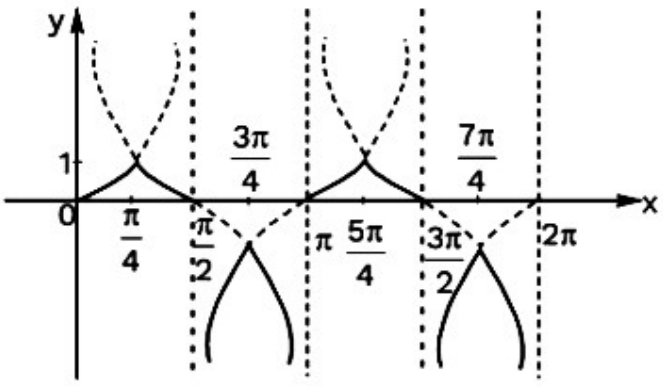
$$=\lim_{t\to 0}\frac{(f(a+2t)-f(a-2t))}{t}$$

Differentiating numerator and denominator

$$=\lim_{t\to 0}\frac{2f'(a+2t)+2f'(a-2t)}{1}=2f'(a)+2f'(a)=4f'(a)=1$$

Q.5 **Consider the function f(x) = Min(tanx, cotx) in (0, 2π). Number of points where f fails to be derivable is 'm' and number of points where it is discontinuous in 'n'. Find (m, n)**

Sol. Number of points where f is non derivable = 7.



Number of points where f(x) is discontinuous = 3

(m, n) = (7, 3).

Q.6 **Discuss the differentiability of f(x) = |x| + |x−1|**

Sol. f(x) = |x| + |x−1|

f(x) is continuous everywhere at |x| and |x − 1| are continuous for all x.

Also, |x| and |x − 1| are non-differentiable at x = 0 and x = 1, respectively.

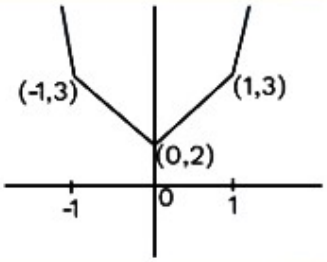
Hence, f(x) is non-differentiable at x = 0 and x = 1.

Q.7 **Discuss the continuity and differentiability of f(x) = |x + 1| + |x| + |x−1|, ∀ x ∈ R; also draw the graph of f(x).**

Sol. f(x) = |x + 1| + |x| + |x−1|, ∀ x ∈ R

Continuous ∀ x ∈ R

Non differentiable at x = −1, x = 0, x = 1



Q.8 **Discuss the differentiability of function f(x) = |[x] x| in -1 ≤ x ≤ 2, where [.] represents greatest integer function.**

Sol. f(x) = |[x] x| $-1 \le x \le 2$

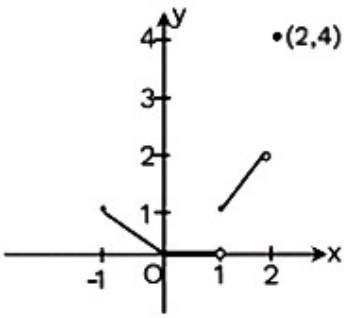
for $-1 \le x < 0$ $f(x) = |-x| = -x$

for $0 \le x < 1$ $f(x) = 0$

for $1 \le x < 2$ $f(x) = |x| = x$

for x = 2 f(x) = |2x| = 2x = 4

x = 1, x = 0, x = 2 are points of non-differentiability.



Q.9 **Discuss the differentiability of f(x) = max {2sinx, 1−cosx} ∀ x∈ (0, π).**

Sol. f(x) = max.{2sinx, 1−cosx} can be plotted as

Thus, f(x) = max {2sinx, 1 − cosx} is not differentiable,

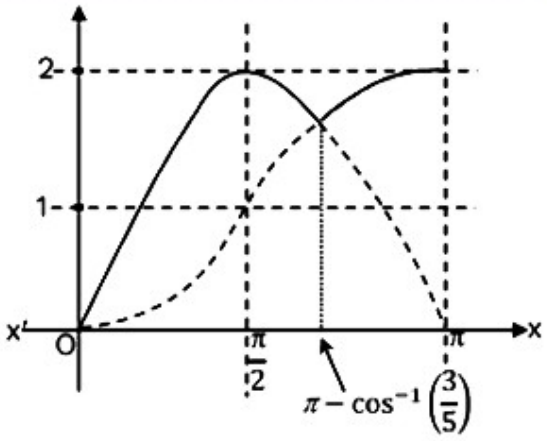
when 2 sinx = 1 − cosx

$\Rightarrow 4\sin^2 x = (1 - \cos x)^2$

$\Rightarrow 4(1 + \cos x) = (1 - \cos x)$

$\Rightarrow 4 + 4\cos x = 1 - \cos x$

$\Rightarrow \cos x = -\frac{3}{5}$



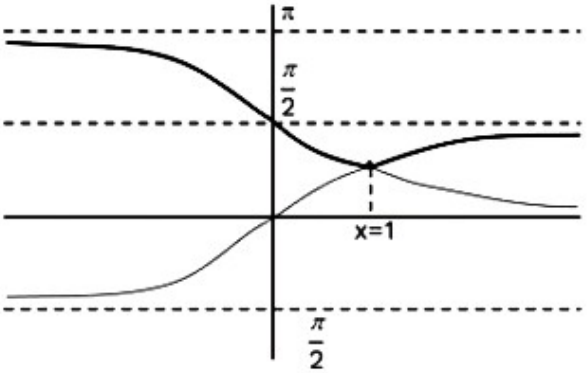
$\Rightarrow \quad x = \cos^{-1}\left(-\frac{3}{5}\right)$

$\Rightarrow$ f(x) is not differentiable at

$x = \pi - \cos^{-1}\left(\frac{3}{5}\right), \ \forall\, x \in (0, \pi)$

Q.10 **Discuss the differentiability of** $f(x) = \max\left\{\tan^{-1} x, \cot^{-1} x\right\}$

Sol. $f(x) = \max.\left\{\tan^{-1} x, \cot^{-1} x\right\}$.

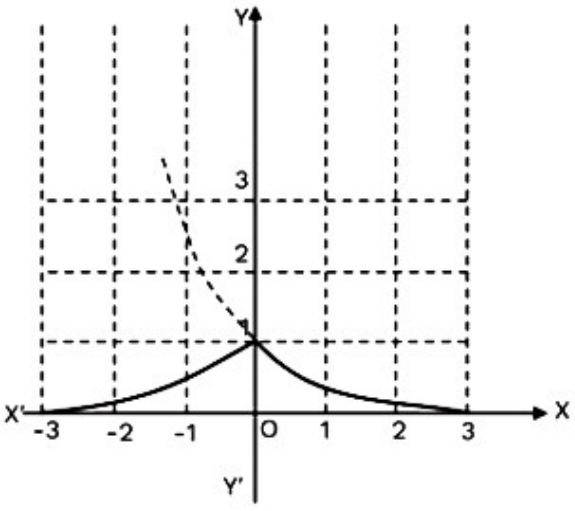


x = 1 is the point of non-differentiability.

Q.11 **Discuss the differentiability of** $f(x) = e^{-|x|}$

Sol. We have

$$f(x) = \begin{cases} e^{-x}, x \geq 0 \\ e^{x}, x < 0 \end{cases}$$



Clearly from the graph,

f(x) is non-differentiable at x = 0.

Q.12 **Discuss the differentiability of** $f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$.

Sol. $f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = y$ (say)

Put $x = \tan\theta$

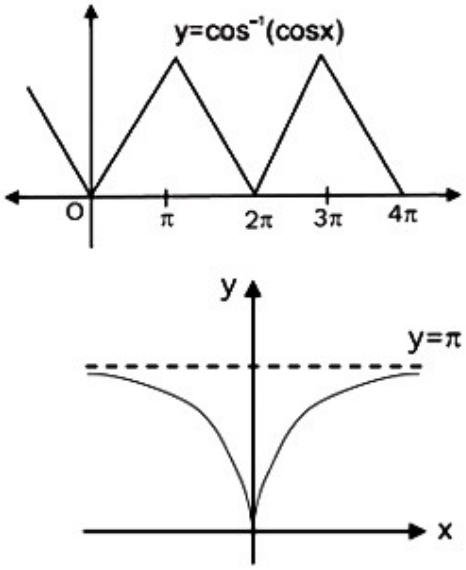
$$y = \begin{cases} 2\theta, & 2\theta \geq 0 \\ -2\theta, & 2\theta < 0 \end{cases}$$

$$f(t) = \cos^{-1}(\cos t)$$

$$y = \begin{cases} 2\tan^{-1} x, & \tan^{-1}x \geq 0 \quad \text{or} \quad x \geq 0 \\ -2\tan^{-1} x, & \tan^{-1}x < 0 \quad \text{or} \quad x < 0 \end{cases}$$

$$y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

f(x) is not differentiable at x = 0.



Determination of function which are differentiable and satisfying the given functional rule :

BASIC STEPS :

(i) Write down the expression for f' (x) as

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

(ii) Manipulate f(x + h) – f(x) in such a way that the given functional rule is applicable. Now apply the functional rule and simplify the RHS to get f'(x) as a function of x along with constants if any.

(iii) Integrate f'(x) get f(x) as a function of x and a constant of integration. In some cases a Differential Equation is formed which can be solved to get f(x).

(iv) Apply the boundary value conditions to determine the value of this constant.

Q.1 **Let f be a differentiable function satisfying $f\left(\frac{x}{y}\right) = f(x) - f(y)$ for all x, y > 0. If f'(1) = 1 then find f(x).**

Sol. $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{1}{h} f\left(\frac{x+h}{x}\right) = \lim_{h\to 0} \frac{1}{h} f\left(1 + \frac{h}{x}\right) = \lim_{h\to 0} \frac{f\left(1+\frac{h}{x}\right) - f(1)}{h}$

as f(1) = 0

So, $f'(x) = \frac{f'(1)}{x} = \frac{1}{x}$

integrating w.r.t. to x

f(x) = lnx + k as f(1) = 0, So k = 0

f(x) = lnx

Q.2 **Suppose f is a derivable function that satisfies the equation $f(x + y) = f(x) + f(y) + x^2y + xy^2$ for all real numbers x and y. Suppose that $\lim_{x\to 0} \frac{f(x)}{x} = 1$, find f(x).**

Sol. $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{f(h) + x^2h + xh^2}{h}$

$f'(x) = \lim_{h\to 0} \frac{f(h)}{h} + x^2 = 1 + x^2$

$f(x) = x + \frac{x^3}{3} + c$ as f(0) = 0 so c = 0

$f(x) = x + \frac{x^3}{3}$

Q.3 **A differentiable function satisfies the relation f(x + y) = f(x) + f(y) + 2xy – 1 $\forall$ x, y $\in$ R. If $f'(0) = \sqrt{3 + a - a^2}$ find f(x) and prove that f(x) > 0 $\forall$ x $\in$ R**

Sol. $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{f(h) + 2xh - 1}{h}$

$f(0) = 2f(0) - 1 \Rightarrow f(0) = 1$

$f'(x) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} + 2x = f'(0) + 2x$

$f'(x) = 2x + \sqrt{(3 + a - a^2)}$

$f(x) = x^2 + \sqrt{(3 + a - a^2)}x + c$

As f(0) = 1 so c = 1

$f(x) = x^2 + \sqrt{(3 + a - a^2)}x + 1$

$D = 3 + a - a^2 - 4 = a - a^2 - 1 = -(a^2 - a + 1) < 0$

as $a^2 - a + 1 > 0 \ \forall \ a$

So, $f(x) > 0 \ \forall \ x \in R$

Q.4 **If f(x + y) = f(x).f(y), ∀x, y ∈ R and f(x) is a differentiable function everywhere. Find f(x).**

Sol. $f(x + y) = f(x)f(y) \Rightarrow f(0) = f(0)^2 \Rightarrow f(0) = 0$ or $f(0) = 1$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{f(x)f(h) - f(x)}{h} = f(x)f'(0)$$

f(0) = 1, assume $f'(0) = k_2$

$$\int \frac{f'(x)}{f(x)} = \int k_2 dx \Rightarrow f(x) = k_1 e^{k_2 x}$$

But $f(0) = 1 \Rightarrow k_1 = 1 \Rightarrow f(x) = e^{k_2 x}$

Q.5 **If f(x + y) = f(x) + f(y), ∀x, y ∈ R then prove that f(ax) = af(x) for ∀ a, x ∈ R.**

Sol. $$f'(x) = \lim_{h \to 0} \frac{f(x+y) - f(x)}{h} = \lim_{h \to 0} \frac{f(x) + f(h) - f(x)}{h} = f'(0) = k$$

$\frac{d}{dx}(f(x)) = k \Rightarrow f(x) = kx + c$, as $f(0) = 0 \Rightarrow c = 0$

$f(x) = kx$

$\Rightarrow f(ax) = k(ax)$
$= akx$
$= af(x)$

Hence Proved.

DIFFERENTIABILITY BY DIFFERENTIATION :

Q.1 **If** $f(x)=\begin{cases} x, & x\le 1 \\ x^2+bx+c, & x>1 \end{cases}$, **then find the values of b and c if f(x) is differentiable at x = 1.**

Sol. $f(x)=\begin{cases} x, & x\le 1 \\ x^2+bx+c, & x>1 \end{cases} \Rightarrow f'(x)=\begin{cases} 1, & x<1 \\ 2x+b, & x>1 \end{cases}$

f(x) is differentiable at x = 1
Then, it must be continuous at x = 1

for which $\lim\limits_{x\to 1^+} f(x)=\lim\limits_{x\to 1^-} f(x)$

$\Rightarrow 1+b+c=1$

$\Rightarrow b+c=0$

Also, $f'(1^+)=f'(1^-)$ ...(i)

$\Rightarrow \quad 2+b=1 \Rightarrow b=-1$

$\Rightarrow c=1$ [from equation(i)]

Q.2 **Find the values of a and b if** $f(x)=\begin{cases} a+\sin^{-1}(x+b), & x\ge 1 \\ x, & x<1 \end{cases}$ **is differentiable at x = 1.**

Sol. $f(x)=\begin{cases} a+\sin^{-1}(x+b), & x\ge 1 \\ x, & x<1 \end{cases} \Rightarrow f'(x)=\begin{cases} \dfrac{1}{\sqrt{1-(x+b)^2}}, & x>1 \\ 1, & x<1 \end{cases}$

For f(x) to be continuous at x = 1.

$f(1^+)=f(1^-) \quad \Rightarrow \quad a+\sin^{-1}(1+b)=1$ ...(i)

Also $f'(1^+)=f'(1^-) \Rightarrow \dfrac{1}{\sqrt{1-(1+b)^2}}=1 \Rightarrow b=-1$

$\Rightarrow a=1$ [Using (i)]

Q.3 **Find the value of a and b if** $f(x)=\begin{cases} ax^2+1, & x \le 1 \\ x^2+ax+b, & x>1 \end{cases}$ **is differentiable at x = 1.**

Sol. $f(x)=\begin{cases} ax^2+1, & x \le 1 \\ x^2+ax+b, & x>1 \end{cases}$ is differentiable at x = 1

$$f'(x)=\begin{cases} 2ax, & x<1 \\ 2x+a, & x>1 \end{cases}$$

L.H.D. at x = 1 = R.H.D at x = 1

$2a = 2 + a \Rightarrow a = 2$

f(x) is continuous at x = 1, hence

$L.H.L\,|_{x=1} = R.H.L\,|_{x=1}$

$2 + 1 = 1 + 2 + b \Rightarrow b = 0$

hence a = 2, b = 0